

SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Department of Computer Science and Engineering

CO2 -- ASSESSMENT TOOL 2 (CO2 AT2)

Student Name: Pavithra R
Register Number: 192421369

Course Code: ITA0404
Course Title: STATISTICS WITH R PROGRAMMING

Course Outcome Covered (CO2): Implement looping constructs, control structures, and recursive functions in R to solve computational problems with efficient program logic. Analyse the Correlation between the parameters in a dataset and establish the analysis (K3).

Weightage: 40% | **Total Marks:** 50 | **Duration:** 1 Hour

1. EMPLOYEE BONUS CALCULATOR

Write an R function `bonus ( )` that accepts an employee's salary and performance rating.

- If rating ≥ 4 , give a 10% bonus. Otherwise, give a 5% bonus.
- Use if-else statement. Return the bonus amount. Set default rating as 3.

```
# Function to calculate employee bonus
bonus <- function(salary, rating = 3) {
  if (rating >= 4) {
    bonus_amount <- salary * 0.10
  } else {
    bonus_amount <- salary * 0.05
  }
  return(bonus_amount)
}

# --- Execution & Test Cases ---
cat("--- Employee Bonus Calculations ---\n")
salary_input <- 50000

# Test Case 1 & 2: Rating >= 4 (5 and 4)
cat("Salary: ₹", salary_input, "| Rating: 5 -> Bonus: ₹", bonus(salary_input, 5), "\n")
cat("Salary: ₹", salary_input, "| Rating: 4 -> Bonus: ₹", bonus(salary_input, 4), "\n")

# Test Case 3 & 4: Rating < 4 (2 and Default 3)
cat("Salary: ₹", salary_input, "| Rating: 2 -> Bonus: ₹", bonus(salary_input, 2), "\n")
cat("Salary: ₹", salary_input, "| Default Rating (3) -> Bonus: ₹", bonus(salary_input), "\n")
```

Expected Console Output:

```
--- Employee Bonus Calculations ---
Salary: ₹ 50000 | Rating: 5 -> Bonus: ₹ 5000
Salary: ₹ 50000 | Rating: 4 -> Bonus: ₹ 5000
Salary: ₹ 50000 | Rating: 2 -> Bonus: ₹ 2500
Salary: ₹ 50000 | Default Rating (3) -> Bonus: ₹ 2500
```

2. SHOPPING DISCOUNT CALCULATOR

Write an R function `discount()` that accepts the purchase amount and membership status.

- If amount > ₹5,000 and customer is a member, give a 10% discount. Otherwise, give no discount.
- Use Boolean operators (&&). Return discount amount. Set default membership status as FALSE.

```
# Function to calculate shopping discount
discount <- function(amount, is_member = FALSE) {
  if (amount > 5000 && is_member) {
    discount_amount <- amount * 0.10
  } else {
    discount_amount <- 0
  }
  return(discount_amount)
}

# --- Execution & Test Cases ---
cat("--- Shopping Discount Calculations ---\n")
cat("Amount: ₹6500 | Member: TRUE -> Discount: ₹", discount(6500, TRUE), "\n")
cat("Amount: ₹6500 | Member: FALSE -> Discount: ₹", discount(6500), "\n")
cat("Amount: ₹4000 | Member: TRUE -> Discount: ₹", discount(4000, TRUE), "\n")
```

Expected Console Output:

```
--- Shopping Discount Calculations ---
Amount: ₹6500 | Member: TRUE -> Discount: ₹ 650
Amount: ₹6500 | Member: FALSE -> Discount: ₹ 0
Amount: ₹4000 | Member: TRUE -> Discount: ₹ 0
```

3. LOAN ELIGIBILITY CHECKER

Write an R function `loan_check()` that accepts age and income.

- Eligible if Age ≥ 21 AND Income ≥ ₹30,000. Use if-else and Boolean operators.
- Return "Eligible" or "Not Eligible".

```
# Function to check loan eligibility
loan_check <- function(age, income) {
  if (age >= 21 && income >= 30000) {
    return("Eligible")
  } else {
    return("Not Eligible")
  }
}

# --- Execution & Test Cases ---
cat("--- Loan Eligibility Checks ---\n")
cat("Age: 25 | Income: ₹35,000 -> Status:", loan_check(25, 35000), "\n")
cat("Age: 19 | Income: ₹40,000 -> Status:", loan_check(19, 40000), "\n")
cat("Age: 23 | Income: ₹25,000 -> Status:", loan_check(23, 25000), "\n")
cat("Age: 18 | Income: ₹15,000 -> Status:", loan_check(18, 15000), "\n")
```

Expected Console Output:

```
--- Loan Eligibility Checks ---
Age: 25 | Income: ₹35,000 -> Status: Eligible
Age: 19 | Income: ₹40,000 -> Status: Not Eligible
Age: 23 | Income: ₹25,000 -> Status: Not Eligible
Age: 18 | Income: ₹15,000 -> Status: Not Eligible
```

4. ELECTRICITY BILL CALCULATOR

Write an R function `electricity_bill()` that accepts units consumed.

- Up to 100 units → ₹2/unit. Above 100 units → ₹3/unit (exceeding 100).
- Add default fixed charge of ₹100. Return total bill. If units < 0, return "Invalid Units".

```
# Function to calculate electricity bill
electricity_bill <- function(units, fixed_charge = 100) {
  if (units < 0) return("Invalid Units")

  if (units <= 100) {
    energy_charge <- units * 2
  } else {
    energy_charge <- (100 * 2) + ((units - 100) * 3)
  }
  return(energy_charge + fixed_charge)
}

# --- Execution & Test Cases ---
cat("--- Electricity Bill Calculations ---\n")
cat("Units: 75 | Fixed: ₹100 -> Total Bill: ₹", electricity_bill(75), "\n")
cat("Units: 150 | Fixed: ₹100 -> Total Bill: ₹", electricity_bill(150), "\n")
cat("Units: 150 | Fixed: ₹150 -> Total Bill: ₹", electricity_bill(150, fixed_charge = 150), "\n")
cat("Units: -25 -> Status:", electricity_bill(-25), "\n")
```

Expected Console Output:

```
--- Electricity Bill Calculations ---
Units: 75 | Fixed: ₹100 -> Total Bill: ₹ 250
Units: 150 | Fixed: ₹100 -> Total Bill: ₹ 450
Units: 150 | Fixed: ₹150 -> Total Bill: ₹ 500
Units: -25 -> Status: Invalid Units
```

5. ATM WITHDRAWAL CHECKER

Write an R function `withdraw()` that accepts balance and amount.

- Check amount > 0 and amount ≤ balance. If true, return remaining balance.
- Otherwise, return "Transaction Failed". Default withdrawal amount: 1000.

```
# Function to process ATM withdrawal
withdraw <- function(balance, amount = 1000) {
  if (amount > 0 && amount <= balance) {
    return(balance - amount)
  } else {
    return("Transaction Failed")
  }
}

# --- Execution & Test Cases ---
cat("--- ATM Withdrawal Checker ---\n")
initial_bal <- 5000
cat("Balance: ₹", initial_bal, "| Default Amount: ₹1000 -> Remaining Balance: ₹",
    withdraw(initial_bal), "\n")
cat("Balance: ₹", initial_bal, "| Amount: ₹2500 -> Remaining Balance: ₹", withdraw(initial_bal,
    2500), "\n")
cat("Balance: ₹", initial_bal, "| Amount: ₹7000 -> Status:", withdraw(initial_bal, 7000), "\n")
cat("Balance: ₹", initial_bal, "| Amount: -₹500 -> Status:", withdraw(initial_bal, -500), "\n")
```

Expected Console Output:

```
--- ATM Withdrawal Checker ---
Balance: ₹ 5000 | Default Amount: ₹1000 -> Remaining Balance: ₹ 4000
Balance: ₹ 5000 | Amount: ₹2500 -> Remaining Balance: ₹ 2500
Balance: ₹ 5000 | Amount: ₹7000 -> Status: Transaction Failed
Balance: ₹ 5000 | Amount: -₹500 -> Status: Transaction Failed
```